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Numerical study on performance of double-fluid parabolic trough solar collector occupied with hybrid non-Newtonian nanofluids: Investigation of effects of helical absorber tube using deep learning

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ABSTRACT

Influences of a helical absorber tube on the thermal performances of a double-fluid parabolic trough solar collector (PTSC) occupied by non-Newtonian nanofluid are investigated numerically. The problem geometry includes a double-fluid parabolic trough solar receiver with high insulation and a non-circular adsorbent tube. The finite volume method, SIMPLE algorithm, and k-ε model were used for numerical solutions. The result showed that the double-fluid solar receiver was equipped with a helical absorber tube filled with nanofluid with a volume fraction of 4% and nanoparticles diameter of 50nm flowing in the absorber tube, and an annulus glass cover is introduced as the optimal model in this paper. For single-flow PTSC, the value of η_{max} nanofluid- and water-based PTSCs are 47.5% and 43.1%, respectively, at a flow velocity of Re=5000. In double-flow PTSC, the value of η_{max} at 1 and 4 vol.% is 56.5% and 58.2%, respectively, at the flow velocity of Re=5000. Finally, a deep learning method is employed to examine the parameters affecting the efficiency, including the concentration and diameter of nanoparticles and Re number. The results indicate the ANN has acceptable performance in prediction of the efficiency of the double-fluid parabolic trough solar collector occupied by non-Newtonian nanofluid.

1. Introduction

Recent estimates via the International Energy Agency (IEA) show that global energy demand will rise by 35% between 2008 and 2035 [1]. Due to the limited fossil energy resources and their disadvantages in the environmental cycle, exploration to find renewable energies to satisfy this growing demand is necessary. The IEA forecasts that more than 13% of the growth in energy demand will be provided by renewable energies [1]. Solar energy is recognized as the purest, cheapest and most available energy [2–8]. Flat-plate solar collectors have a simple design and low manufacturing cost than other types of collectors and are able to absorb scattered radiation in addition to direct solar radiation [9–14]. Hot air and water are widely used in industry, agriculture, animal husbandry and home affairs [15]. Solar collectors can therefore be used to heat the air and water at the same time [16, 17]. With the advancement of nanoscience, scientists improved the properties of various materials. They added nanoparticles to various materials, including solids and liquids, and observed that most of them improved mechanical and thermophysical properties [18–27]. Therefore, by adding nanoparticles to water and other common liquids, researchers could help to improve the efficiency of heating and cooling systems [28–33]. The focus of the present work is on dual-purpose and air-heating solar collectors.

Collectors are devices that convert solar radiation into heat energy by trapping it. They are separated into two common categories, concentrating and non-concentrating. Concentrating collectors consist of parabolas that reflect sunlight at their focal point. Solar radiation can be absorbed by putting these parabolas together and moving fluid from their focal point to use for heating or generation of electricity [34, 35]. In general, the flat-plate collectors and evacuated tube ones can be called a subset of non-concentrating collectors [36–39].

Jebasingh and Herbert [1], investigated a Parabolic Trough Solar Collector (PTSC) in a review paper. They also evaluated the many applications of PTSCs, including air and water heating systems, solar desalination, solar cooling systems, hot steam generation, and other thermal applications industries and power plants. It should be mentioned that their main focus was on solar energy applications in India. According to their study, PTSCs can produce hot steam up to

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Nomenclature

Symbols

c_P	Specific heat, (J/kgK)
D_h	Hydraulic diameter, (m)
D	Diameter of nanoparticles (nm)
F	Friction factor
k	Thermal conductivity, (W/mK)
Pr	Prandtl number
P	Pressure, (Pa)
Re	Reynolds number
T	Temperature (K)
U	Velocity *(m/s)*
V_m	Velocity *(m/s)*

Greek Symbols

α	Thermal diffusion m^2/s
μ	Dynamic viscosity mPa.s
ρ	Density m^3/kg
ϕ	Nanoparticles volume fraction vol.%

Subscriptions

Bf	Base fluid
Nf	Nanofluid
np	Nanoparticle

Abbreviatio

HNFs	Hybrid nanofluid
HTF	HeatTransfer Fluids
PTSC	Parabolic Trough Solar Collector

400°C.

Chafie et al. [40] in 2017 examined energy and exergy efficiency of PTSCs by performing experimental tests. Their main goal was to improve η_{PTSC}. They accurately calculated the useful energy received, the exergy transferred to the working fluid, the energy efficiency, the exergy efficiency, and the exergy factor. Their results showed that $\eta_{collectors}$ depends strongly on the climatic conditions of the collector installation site. Based on their results, the best energy and exergy efficiency occurs on clear sky days. The ranges of the variations of energy and exergy collector efficiencies are 19.7% to 52.6% and 8.5% to 16.3%, respectively. In 2017, Conrado et al. [41] evaluated η_{PTSC}. Their work examined a comprehensive set of investigational and numerical studies in the field of PTSCs. According to their results, the thermodynamics of solar collectors has a significant role in their thermal performance. Based on their results, the researchers' views on the $\eta_{collectors}$include:

- Collecting and combining various thermal models provided by previous authors
- Introduce new numerical methods and simulations with the maximum accuracy and minimum computational requirements
- Proposition of new methods for accurately measuring the thermal efficiency of solar collectors
- Reduce the cost of manufacturing high-efficiency solar collectors

Kumaresan et al. [42], in 2017, investigated numerically and experimentally the thermal performance improvement of PTSCs. Based on their results, PTSCs have been documented as the most optimum solar receivers for medium temperatures working between 160°C and 390°C. Consumers also consider PTSCs due to their high efficiency.

Rohani et al. [43] simulated PTSCs using experimental data. They assessed the impact of different weather conditions throughout the year. Their main goal was to achieve an optimal performance method, including start time and activity period. They validated their simulation results with the experimental results and found that the simulation results have 2.29% error compared to the experimental results. Finally, they provided optimal performance conditions for the collector. In 2018, Abdessemed et al. [44] experimentally examined the influence of tray shape on the performance of the dependent distillers of multi-stage PTSCs. They studied two diverse procedures of V and Λ. According to their results, the use of V reduces costs because the use of Λ system requires two solar collectors. In a review study in 2018, Bellos et al. [45] investigated $\eta_{collectors}$ using nanofluids and turbulators. According to their study, the use of nanofluids and turbulators (for example, the usage of fins in the absorber tube) are the most commonly used methods to improve η_{PTSC}. However, these methods result in a pressure drop. Also, by summarizing the results of others' work, they simulated numerically a PTSC equipped with rectangular fins inside an absorber tube containing CuO/oil nanofluid with volume fraction of 6%. According to their results, the parameters of using nanofluid, rectangular turbulators and simultaneous application of nanofluid and turbulators increased the effectiveness by 0.76%, 1.10% and 1.54%, respectively.

Table 1
List of different numerical studies performed in the field of PTSCs

Description	Thermal model	Numerical method	Computational software
Simple structure	Heat Transfer Model	–	Engineering equations solver
Flexibility High accuracy in thermodynamic calculations Ability to optimize univariate and bivariate Very high computational speed	Energy Model		
SIMPLE algorithm	Heat Transfer Model	FVM	Computational Fluid Dynamics (CFD)
Using discretization methods	Energy Model	FEM	ANSYS CFD
High accurate data	3D Model	Genetic Algorithm	
Precision Identify critical processes High capacity The ability of complex and advanced modeling	Nonuniform Model		
The main goal is to create commercial software that has an open-source platform that can be modified and extended to optimize and simulate Modelica models.	Energy Equation	Finite Difference Methods	Modelica

In a review study in 2019, Bellos and Tzivanidis [46] examined alternative designs for PTSCs. The PTSCs are very efficient for both high-temperature and medium-temperature applications based on their work. However, flat panel collectors are recommended for low-temperature applications. They noted that all research in the field of PTSCs aims to increase thermal efficiency and reduce fabrication costs. Therefore, a lot of research has been done on the geometrical optimization and evaluation of the use of nanofluids. Carballo et al. [47] achieved the energy and exergy analysis of PTSCs in 2019. Like other researchers, their main goal is to increase collector efficiency by reducing construction costs. They prepared a dynamic model for solar collectors' transient flow (time-dependent) and optimized the highest thermal and hydraulic performance evaluation criteria (PEC) and the

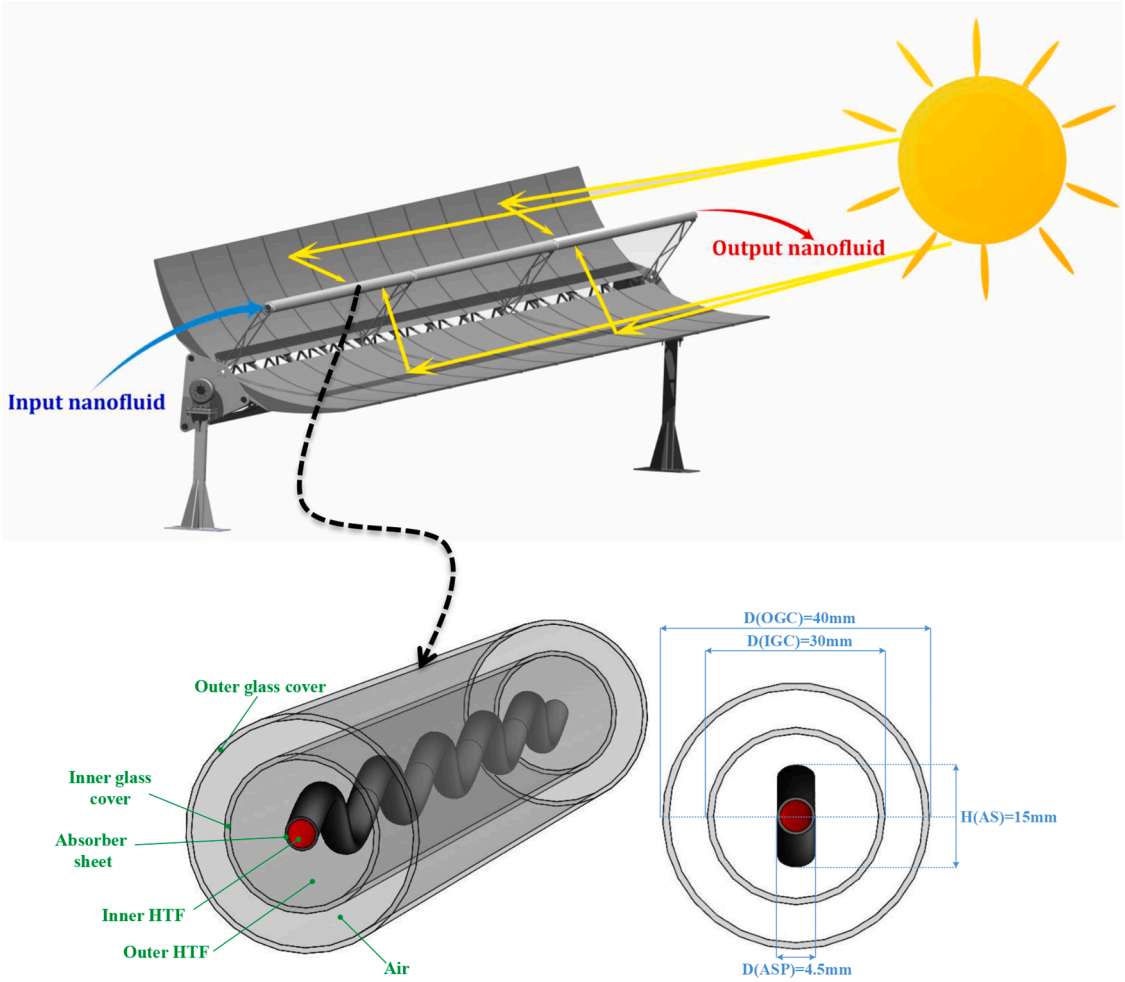


Fig. 1. Geometrical parameter of studied PTSC.

lowest exergy generation. They also validated their new model with previous models and assured the accuracy of their model performance. In 2019, Xu et al. [48] developed a new analytical model that included all the parameters affecting the collector operating conditions such as inlet temperature and inlet flow rate, incident radiation, internal perturbations, optical efficiency and time variations.

Table 1 presents the list of various numerical studies in the field of PTSCs. As shown in this table, different types of software are able to simulate PTSCs using different methods and features. Although engineering equation solvers are very flexible and highly accurate, they have many limitations in complex and three-dimensional geometries. They can only simulate three-dimensional geometries assuming extensive simplifications [49–52]. Another approach is the use of commercial fluid engineering software such as ANSYS FLUENT, CFX, ABAQUS, SolidWorks, COMSOL Multiphysics, and CATIA [53–70]. In the present study, ANSYS FLUENT was used. According to previous research, this software is the most accurate engineering software for simulation of 3D physical models and phenomena and validates numerical results with experimental data.

The ultimate aim of all numerical research, including research on PTSCs, is to achieve a model that can improve collector performance under real conditions. According to the work of Conrado et al. [41], the performance improvement of PTSCs is summarized as follows:

- Reduction of fabrication costs (economical due to collector performance)
- Reduction of heat loss of the collector
- Improvement of collector heat transfer efficiency
- Reduction of pressure drop inside the collector

But, the ideas that will improve $\eta_{collectors}$ are only valuable if they can be implemented at a reasonable cost and time, obviously.

Therefore, it is very important to know the process of fabricating PTSCs and the problems in this way. Thus, researchers who work on PTSCs should first know these problems and then propose methods to improve η_{PTSC}. Conrado et al. [41] carefully examined and presented all of these cases.

As seen in the review of previous work, numerical, analytical, and statistical methods can help researchers solve engineering problems. Their results show that the accuracy of these methods is very good and satisfies the problems well [71–75]. Meanwhile, with the invention of mathematical methods, artificial intelligence approaches, and supercomputers, researchers have also considered these methods to predict the behavior of systems [76–82]. Therefore, in this research, a solar collector is analyzed using numerical and artificial intelligence methods, which can be trusted well. The current investigation aims to attain an optimum configuration for PTSCs with the viewpoints of energy efficiency. In this investigation, different influences of utilizing nanofluid is a novel PTSC employing single- and Two-Phase Procedures (TPP) has been analyzed. The chief innovations and aims of this investigation can be written as follow:

i TPP simulation of non-Newtonian CNT/Fe_3O_4/H_2O/CMC hybrid nanofluid flow in a PTSC prepared with helical tube and to compare the novel configuration with conventional PTSC at similar hydraulic diameter.

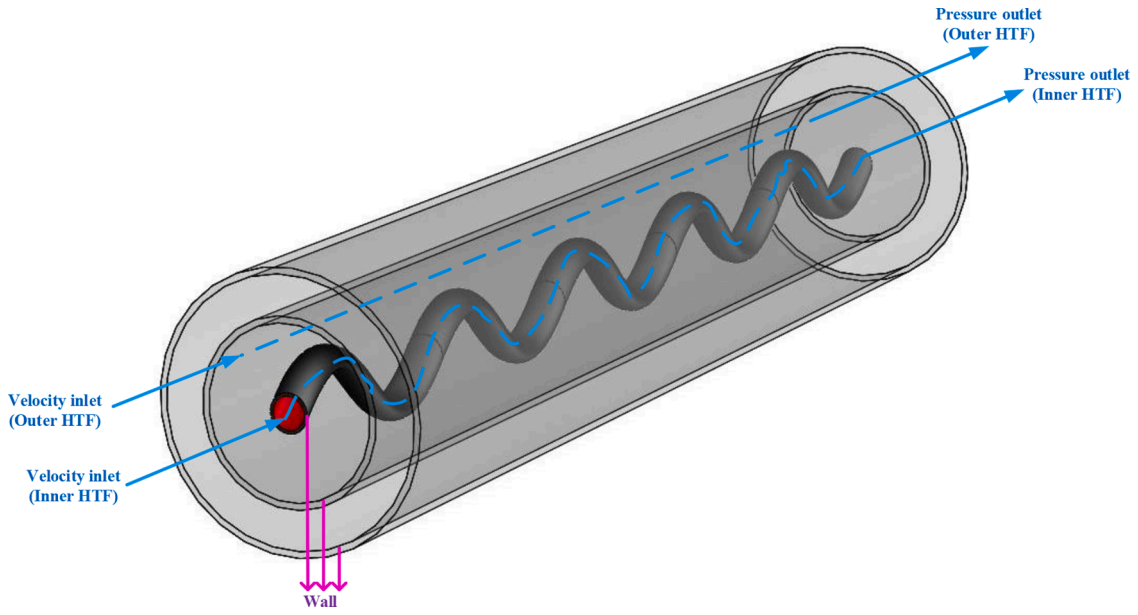


Fig. 2. Schematic view and boundary conditions of the studied problem.

Table 2
Properties of studied nanofluids [83]

Properties	φ_{CNT} = 0.10% φ_M (%)					φ_{CNT} = 1.35% φ_M (%)				
	0.10	0.30	0.50	0.70	0.90	0.10	0.30	0.50	0.70	0.90
$\mu_{nf} \times 10^{-4}\left(\frac{kg}{m.s}\right)$	8.15	9.48	11.08	12.81	14.48	11.33	13.03	14.61	15.95	17.01
$k_{nf}\left(\frac{W}{m.K}\right)$	0.691	0.725	0.739	0.759	0.794	0.703	0.759	0.772	0.866	0.902
	P (kg/m^3) d_{np} = 100nm	c_p (J/kg•K) d_{np} = 45nm	$k\left(\frac{W}{m.K}\right)$ d_{np} = 25nm		Nanofluid CMC (0.5%) + 1.0% + Hybrid CNT 1.35% / M 0.9%					
0.6157	0.6211	0.6262	4121	1013.5	CMC (0.5%) + 1.5% + Hybrid CNT 1.35% / M 0.9%					
0.6157	0.6211	0.6262	4012	1040.5						

ii Effects of using double-flow PTSC on energy efficiency of PTSC equipped with the helical tube.
iii Effect of nano-scale particles concentration on η_{PTSC}

2. Numerical modeling

2.1. Physical model

As stated earlier, the current analysis aims to attain an optimum configuration for PTSCs. In instruction to simulate the PTSC in the present investigation, the present empirical information of an example town has been determined. The problem configuration includes a double-fluid PTSC (DFPTSC) with high insulation and a non-circular adsorbent tube. Fig. 1 demonstrates the present problem configuration. Fig. 2 presents the three-dimensional schematic schema of studied PTSC and also the boundary conditions.

These assumptions are considered in the present study:

1 Nanofluid properties depend on nanoparticles Brownian velocity, temperature and concentration.
2 Nanofluid is modeled using a two-phase approach
3 Simultaneously resolution of the equations convection, conduction and radiation
4 Steady-state solution

The nanofluid engaged in the current investigation is non-Newtonian base fluid water/Carboxymethyl cellulose (CMC) non-Newtonian base fluid (99.5% H_2O: 0.5% CMC) containing Fe_3O_4/CNT nanoparticles with nanoparticle volume concentrations between 0.00 and 1.35 % and nanoparticle size between 25 nm to100 nm. Table 2 demonstrates the nanofluid properties. According to Figure 1, a tube with a helical duct is placed in the middle of a parabolic collector. There is a nanofluid flow inside this tube. The heat radiated to the parabolic collector is reflected on the tube, and the sun's heat is transferred to the nanofluid as the working fluid. Two glass covers prevent heat transfer and prevent the reflection of heat radiated to the collector tube. There is a second fluid between two glass covers, which helps absorb more heat and, on the other hand, reduces heat dissipation from the nanofluid to the surrounding environment.

2.2. Governing Equations

Due to simulate the Fe_3O_4 / CNT + 0.995H_2O: 0.005 CMC HNFs flow inside the studied PTSC; two diverse approaches are determined in this paper. The first one, engaged to the code validation section and also for modeling of air in the annulus, is the *Single-Phase Model* (SPM), which

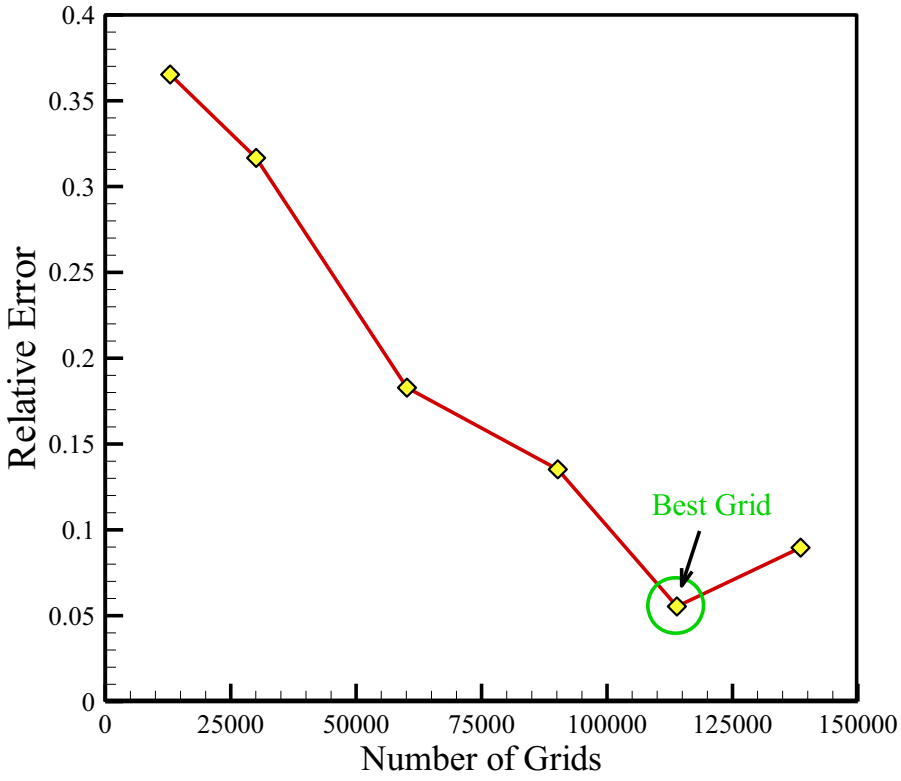


Fig. 3. Grid study.

determines that the solid nanoparticles and fluid phase have the similar temperature and velocity field. The second technique is called Two-Phase Model (TPM) [84–86]. This procedure gives influential estimates for nanofluid simulation in low nanoparticle concentrations [87]. The governing equations for simulation of non-Newtonian nanofluid flow have been utilized [88, 89].

The equation of continuity:

$$\nabla\left(\rho_m \vec{U}_m\right) = 0 \tag{1}$$

where $\vec{U}_m$ is the nanofluid or mass-averaged velocity:

$$\vec{U}_m = \frac{\rho_s \phi_s \vec{U}_s + \rho_{bf}\phi_{bf}\vec{U}_{bf}}{\rho_m} \tag{2}$$

where $\vec{U}_s$ is the velocity of solid nanoparticles and ρ_m is obtained using Eq. (3):

$$\rho_m = \rho_s \phi_s + \rho_{bf}\phi_{bf} \tag{3}$$

The steady-state momentum equation is:

$$\rho_m\left(\vec{U}_m \nabla \vec{U}_m\right) = -\nabla P + \mu_m\left(\nabla \vec{U}_m + \left(\nabla \vec{U}_m\right)^T\right) + \nabla\left(\rho_{bf}\phi_{bf}\vec{U}_{dr,bf}\vec{U}_{dr,bf} + \rho_s\phi_s\vec{U}_{dr,s}\vec{U}_{dr,s}\right) + \rho_m \vec{g} \tag{4}$$

where, P is the pressure, $\vec{U}_{dr,s}$ and $\vec{U}_{dr,bf}$ are drift velocities:

$$\vec{U}_{dr,bf} = \vec{U}_{bf} - \vec{U}_m \tag{5}$$

$$\vec{U}_{dr,s} = \vec{U}_s - \vec{U}_m \tag{6}$$

The energy equation:

$$\nabla\left(\rho_{bf}\phi_{bf}\vec{U}_{bf}h_{bf} + \rho_s\phi_s\vec{U}_s h_s\right) = \nabla\left(\left(\phi_{bf}k_{bf} + \phi_s k_s\right)\nabla\vec{T}\right) \tag{7}$$

where, h_s and h_{bf} are the enthalpy. The volume fraction equations for two-phase nanofluid are [88]:

$$\nabla\left(\rho_s\phi_s\vec{U}_m\right) = -\nabla\left(\rho_s\phi_s\vec{U}_{dr,s}\right) \tag{8}$$

The slip velocity Eq as:

$$\vec{U}_{bf,s} = \vec{U}_{bf} - \vec{U}_s \tag{9}$$

$$\vec{U}_{dr,s} = \vec{U}_{s,bf} - \frac{\rho_s\phi_s}{\rho_m}\vec{U}_{bf,s} \tag{10}$$

According to the Schiller and Naumann [88, 90] drag formulation, the relative velocity can be written as:

$$\vec{U}_{bf,s} = \frac{d_p^2}{18\mu_{bf}f_d}\frac{\rho_s - \rho_m}{\rho_s}\vec{\alpha} \tag{11}$$

$$f_d = 1 + 0.15\mathrm{Re}_s^{0.687} \tag{12}$$

$$\vec{\alpha} = \vec{g} - \left(\vec{U}_m \nabla \vec{U}_m\right) \tag{13}$$

The Reynolds number of nanoparticles (Re_s) can be written as:

$$\mathrm{Re}_s = \frac{\vec{U}_m d_p \rho_m}{\mu_m} \tag{14}$$

where d_p is the average nanoparticles diameter.

The flow in the absorber tube is considered turbulent, and k-ε model has also been used [85]. The choice of the turbulence k-ε model is because of its wide validation. The turbulence k-ε model has been well utilized in various relevant mathematical studies in the field of PTSC

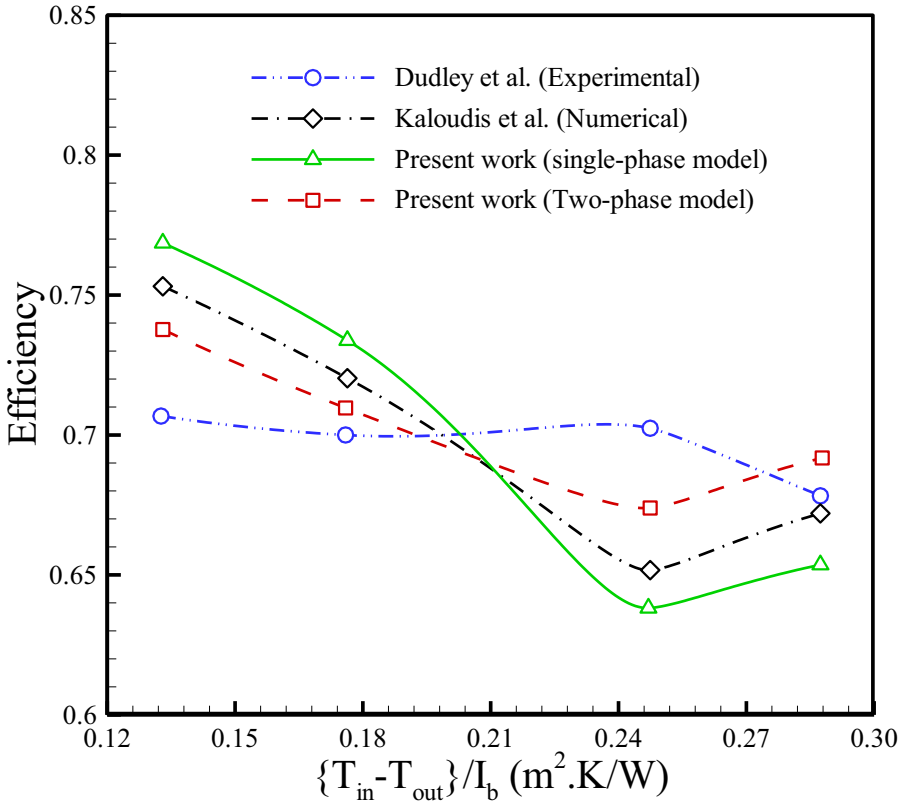


Fig. 4. The validation among numerical data achieved from present study, the investigational data by [91] and also the numerical data obtained by [92] in case of employing single- and two-phase methods for simulation of the non-Newtonian hybrid nanofluid.

simulation. The relations which define the k-ε model are as follows:

$$\nabla\left(\rho_m \vec{U}_m k\right) = \nabla\left[\left(\mu_m + \frac{\mu_{t,m}}{\sigma_k}\right)\nabla k\right] + G_{k,m} - \rho_m \varepsilon \tag{15}$$

$$\nabla\left(\rho_m \vec{U}_m \varepsilon\right) = \nabla\left[\left(\mu_m + \frac{\mu_{t,m}}{\sigma_\varepsilon}\right)\nabla \varepsilon\right] + \frac{\varepsilon}{k}\left(c_1 G_{k,m} - c_2 \rho_m \varepsilon\right) \tag{16}$$

where G is the production rate of k and $\mu_{t,m}$ is the turbulent viscosity:

$$\mu_{t,m} = C_\mu \rho_m \frac{k^2}{\varepsilon} \tag{17}$$

$$G_{k,m} = \mu_{t,m}\left(\nabla \vec{U}_m + \left(\nabla \vec{U}_m\right)^T\right) \tag{18}$$

The standard constants are employed, $C_\mu = 0.09$, $c_1 = 1.44$, $c_2 = 1.92$, $\sigma_k = 1.00$, $\sigma_\varepsilon = 1.30$ and $\sigma_t = 0.85$. The radiative heat transfer was modeled by the Monte Carlo approach [85].

The RMS residual was less than 10^{-6} for all heat transfer variables and nanofluid flow in PTSC in the converging standard values.

For analysis of the characteristics of nanofluid flow and heat transfer qualifications in studied PTSCs, some beneficially interested parameters have been listed in this section.

Reynolds number:

$\mathrm{Re} = \frac{\rho_{bf} \cdot u_m \cdot d_a}{\mu_{bf}}$ (19) where u_m is the mean velocity of nanofluid inside the PTSC. The parameter of η_c could be calculated using Eq. (20). For the new double-fluid PTSC, although the energy efficiency η_n may be written utilizing Eq. (21):

$$\eta_c = \frac{E_c}{I \cdot A} = \frac{Q_{in} \cdot \rho_{in} \cdot c_{p,in} \cdot (T_{out} - T_{in})}{6 \cdot 10^4 \cdot I \cdot A} \tag{20}$$

$$\eta_n = \frac{E_n}{I \cdot A} = \frac{p_{fo} \cdot Q_{in} \cdot \rho_{in} \cdot c_{p,in} \cdot (T_{fo,out} - T_{in}) + (1 - p_{fo}) Q_{in} \cdot \rho_{in} \cdot c_{p,in} \cdot (T_{fi,out} - T_{in})}{6 \cdot 10^4 \cdot I \cdot A} \tag{21}$$

2.3. Grid Independency

The grid independence test was done to analyze the impacts of grid mesh size on the accuracy of consequences. As it is seen in Fig. 3, different grid meshes have been generated and analyzed for the studied PTSC. By comparing these different grid meshes, it is realized that in grid mesh size of 113453, the minimum error values are achieved by Nusselt number calculations; thus, this grid is selected as the best model with the maximum accuracy minimum time costs.

2.4. Validation

Fig. 4 shows the comparison among numerical data achieved from the present study, the investigational data of Dudley et al. [91] and also the numerical data achieved by the study of Claudius et al. [92]. As it is seen in this figure, there is a suitable coincidence among the obtained results from the present study, the investigational data of Dudley et al. [91] and also the numerical data found from the study of Claudius et al. [92] with the maximum error of 10.98%.

3. Results and discussion

In the current part, first, the influences of using a PTSC system equipped with conventional and helical absorber tubes are investigated. Then the effects of using single and two-fluid systems are investigated. Furthermore, the influence of using nanofluid on η_{PTSC} has been investigated. Finally, the two-phase non-Newtonian hybrid nanofluid

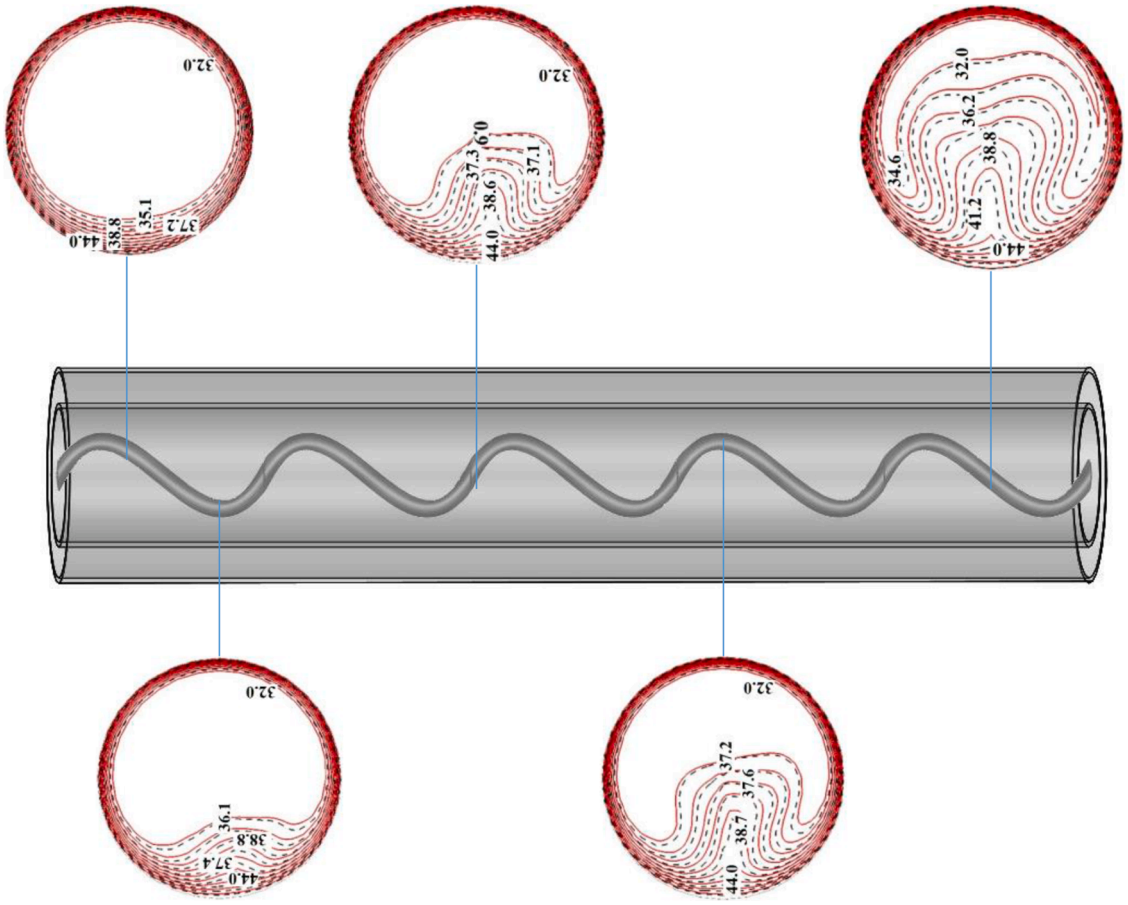


Fig. 5. The isothermal lines in the double-fluid PTSC filled with base fluid for the helical absorbent channel.

properties, including the nanoparticles size and volume fraction (for the fluid inside the adsorbent tube and within the glass cover), are investigated.

Fig. 5 shows the isothermal lines in the two-fluid PTSC filled with base fluid for the helical absorbent channel. Fig. 6 also demonstrates the isothermal lines in a double-fluid PTSC filled with nanofluid for the helical absorbent channel. Isothermal lines are a tool for a qualitative study of the coolant behavior in the collector duct. The curvature of the isothermal lines indicates a greater convection coefficient of nanofluid in the duct and, consequently, a higher heat transfer rate. The greater density of isothermal lines also indicates a more intense temperature gradient in those areas and, as a result, a greater heat exchange rate. It is realized that the thermal diffusion of heat into the fluid inside the adsorbent tube is significantly increased by using nanofluid.

Fig. 7 demonstrates the first law efficiency variations of a single-flow PTSC filled with water versus different Re for conventional and helical absorber tubes models with the same hydraulic diameter. Different effects of using two types of straight and helical adsorbent tubes in a single-flow (water) system have been investigated. As can be seen, for both conventional and helical tubes, the energy efficiency always grows by an increase in Re. Hence, there is an optimum flow velocity (Re = 5000) in which the maximum energy efficiency values are attained. Based on Fig. 7, the first law efficiency values for the novel PTSC are always higher than that of the conventional PTSC during all studied Re. Clearly, this behavior is because of the longer length of stream in the novel PTSC. Also, in the helical tube, the absorption of solar energy from the reflector increases due to the greater surface area of heat transfer. The helical system also has a higher flow rate at the same Re, and thus the heat transfer rate from channel walls to HTF flow increases. Visibly, the thermal diffusion depth inside the HTF reduces for larger diameters of the absorber tube. Also, the occupied presence of helical-shaped sections in the flow path causes local turbulence in HTF flow. Also, it could break the laminar sub-layers and increase the heat transfer rate inside the helical scheme. Instead, the radiation received is concerted only on the lower section of the conventional tube, while it is concentrated on lower and absorbent channel side parts in the helical case because of the special configuration of the absorbent channel. The highest values of energy efficiency for the helical tube and conventional one at the Reynolds number of 5000 are 42.95% and 38.88%, respectively. Therefore, it is realized that usage of a helical tube for the single-fluid PTSC, could enlarge the energy efficiency value of the collector. Consequently, trendy the present work, this helical absorber tube system is used to investigate other parameters.

Fig. 8 illustrates the variations of energy efficiency values versus different Reynolds numbers for a single-fluid PTSC prepared with helical absorber tube system occupied by base fluid and nanofluid ($\phi = 1\%$ and $d_{np} = 20$nm). As it is understood in this figure, occupying of nanofluid has a significant influence on η_{PTSC} and the energy efficiency values of the single-fluid PTSC filled with nanofluid are always more than the energy efficiency values of the single-fluid PTSC filled with water during all studied Reynolds numbers. Usage of nanofluid leads to more dynamic viscosity in HTF and hence could rise the pressure drop. Also, it is seen that for both studied models in this figure, the energy efficiency values always increase with the rise of the Reynolds number. Therefore, it can be assumed that there will be η_{max} on $V_{optimum}$. The value of η_{max} for nanofluid- and water-based PTSCs are 47.5% and 43.1%, respectively, at flow velocity of Re= 5000.

Fig. 9 illustrates variations of energy efficiency values versus different *Re* for single- and two-fluid PTSCs equipped with helical absorber tube systems filled with nanofluid. Utilizing double-fluid

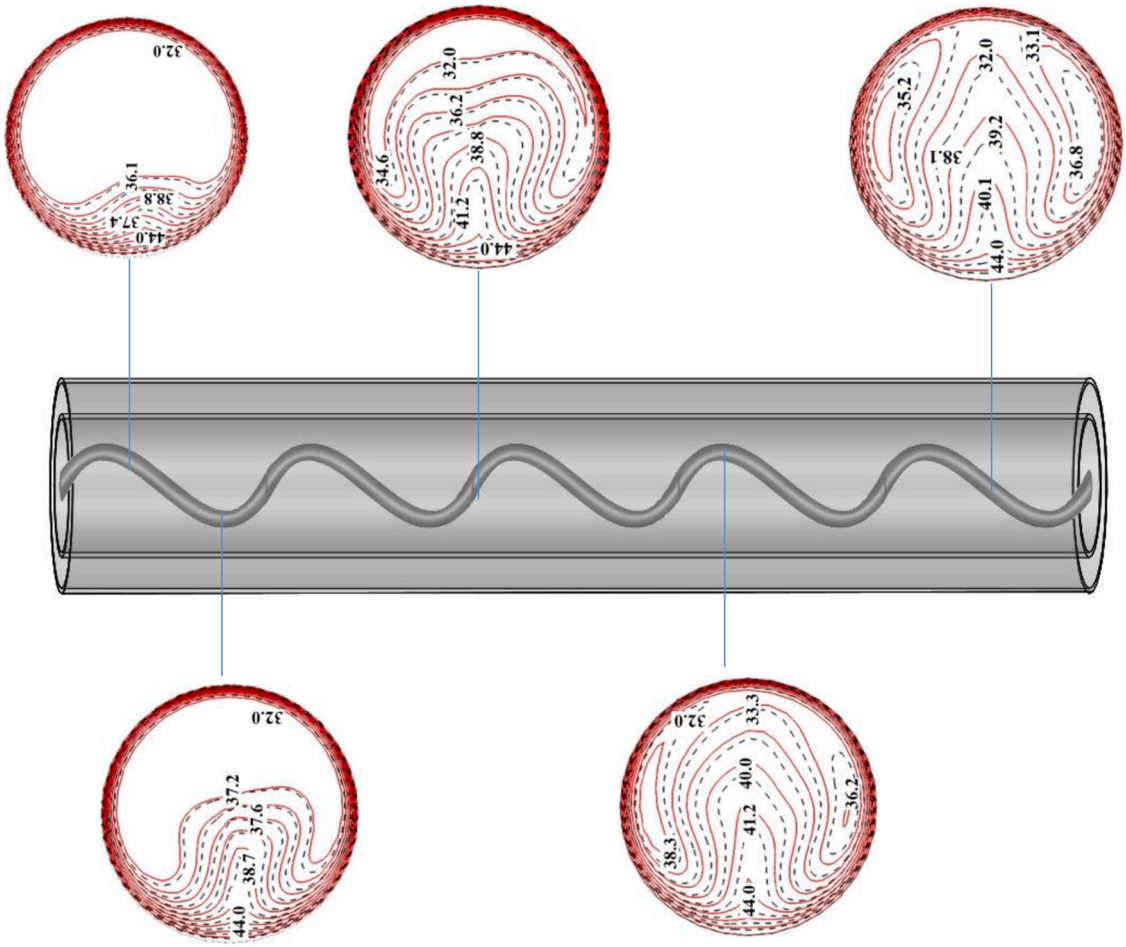


Fig. 6. The isothermal lines in a two-fluid PTSC containing nanofluid for the helical absorber tube system.

system has a significant impact on the energy efficiency enhancement of PTSC. The energy efficiency values of the double-fluid PTSCs filled with nanofluid are always more than the energy efficiency values of the single-fluid PTSC filled with nanofluid during all studied *Re*. Usage of double fluid system leads to more heat transfer area and therefore enhances the heat transfer rate between HTF and absorber tube. Besides, usage of double fluid system can lead to lower heat loss value from the absorber tube. Also, it is seen that for all considered models in this figure, the energy efficiency values always rise with an increase of *Re*. Therefore, it can be assumed that there will be η_{max} on $V_{optimum}$. On the other side, it is understood that the use of nanofluid in both paths, outer annulus and inner tube can be principal to higher heat transfer enhancement in PTSC. The maximum energy efficiency values for the PTSC filled with suspension in an internal tube, and external path (annulus) and the single-HTF PTSC filled with nanofluid are 56.4% and 47.4%, respectively, at flow velocity, resulting in Re = 5000.

Fig. 10 demonstrates differences of the first law efficiency of a double-flow PTSC against varied *Re* equipped with a helical tube system occupied by nanofluid and glass cover annulus occupied with base fluid for various ϕ at a constant diameter. As it can be understood in this figure, use of nanofluid with different nanoparticles volume fractions has a significant influence on energy efficiency enhancement of PTSC. It is seen that the energy efficiency always augmentations with rising the volume fractions. The PTSC engaged with nanofluid at ϕ = 4vol% has the maximum energy efficiency values among all considered cases and is followed with models filled by ϕ = 3, 2 and 1vol% nanofluid, respectively. Increasing volume fraction increases the thermal conductivity of nanofluid into HTF. Therefore enhances $h_{convection}$ between HTF and tube. Besides, increasing volume fraction increases the dynamic viscosity nanofluid into HTF. Therefore raise the pressure drop between inlet and outlet sections. Also, it is seen that for all considered models in this figure, the energy efficiency values always rise with an increase of *Re*. Therefore, it can be assumed that there will be η_{max} on $V_{optimum}$. The highest first law efficiencies for nanoparticles volume concentrations of 1vol% and 4vol% are 53.2% and 54.8%, respectively, at the flow velocity of Re = 5000.

Fig. 11 shows variations of first law efficiency of a double-flow PTSC versus different flow velocities equipped with the helical absorber tube system filled with nanofluid at different nanoparticles volume fractions and constant diameter. As understood in this figure, engaging nanofluid with different nanoparticles volume fractions has a significant influence on the energy efficiency enhancement of PTSC. It is realized that the nanoparticles volume fractions increase, leading to enhanced energy efficiency. The PTSC occupied with ϕ = 4vol% nanofluid has the maximum energy efficiency values amongst all considered cases and is followed with models filled by nanofluid at ϕ = 3, 2 and 1vol%, respectively. A higher volume fraction increases the thermal conductivity of nanofluid in HTF and therefore enhances $h_{convection}$ between HTF and tube. Besides, nanofluid application with higher ϕ leads to more dynamic viscosity in HTF and hence can rise the pressure drop between inlet and outlet sections. Also, it is seen that for all considered models in this figure, the energy efficiency values are augmented by rising of *Re*. Therefore, it can be assumed that there will be η_{max} on $V_{optimum}$. The value of η_{max} at 1 and 4 vol.% is 56.5% and 58.2%, respectively, at the flow velocity of Re = 5000.

Fig. 12 demonstrates variations of energy efficiency of a double-fluid PTSC versus different *Re* fixed with helical absorber tube arrangement engaged by nanofluid at various ϕ with the constant diameter (for

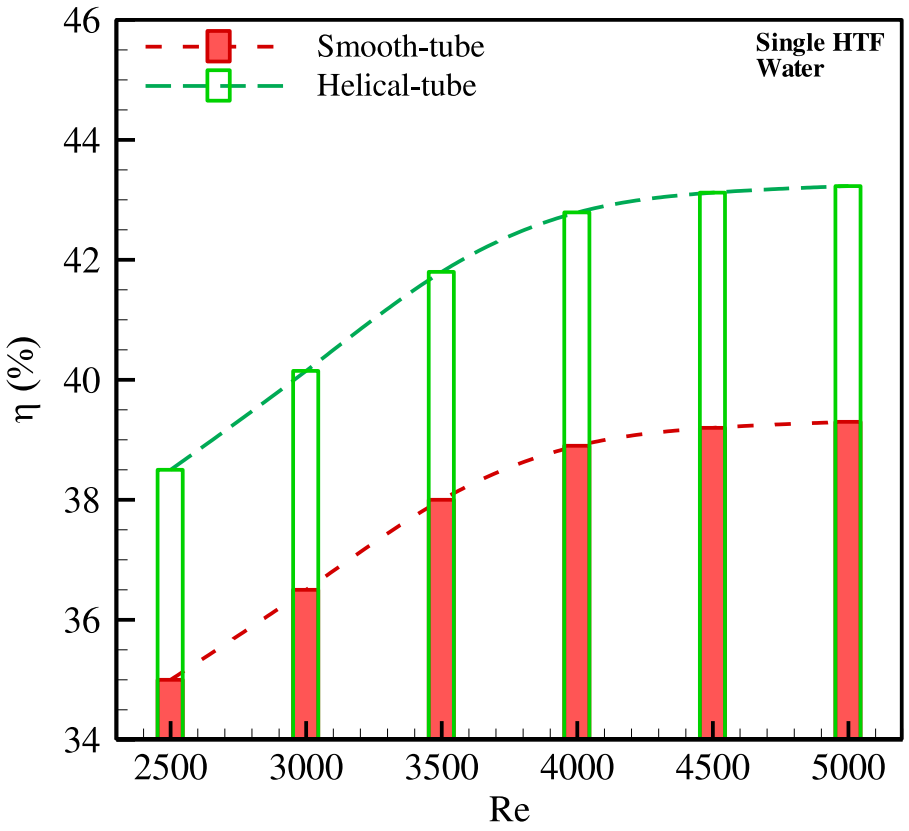


Fig. 7. Variations of energy efficiency value of the single-fluid PTSC containing water versus different Reynolds numbers for conventional and helical absorber tubes with similar hydraulic diameters.

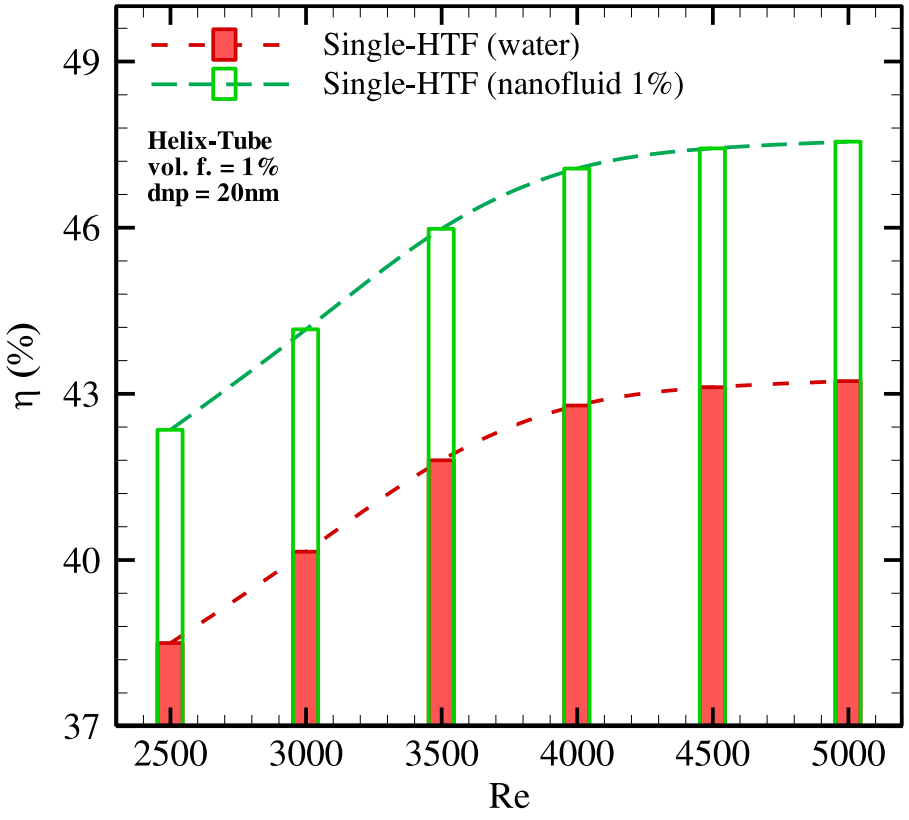


Fig. 8. Variations of energy efficiency values versus different Reynolds numbers for a single-fluid PTSC prepared with helical absorber tube system occupied by base fluid and nanofluid.

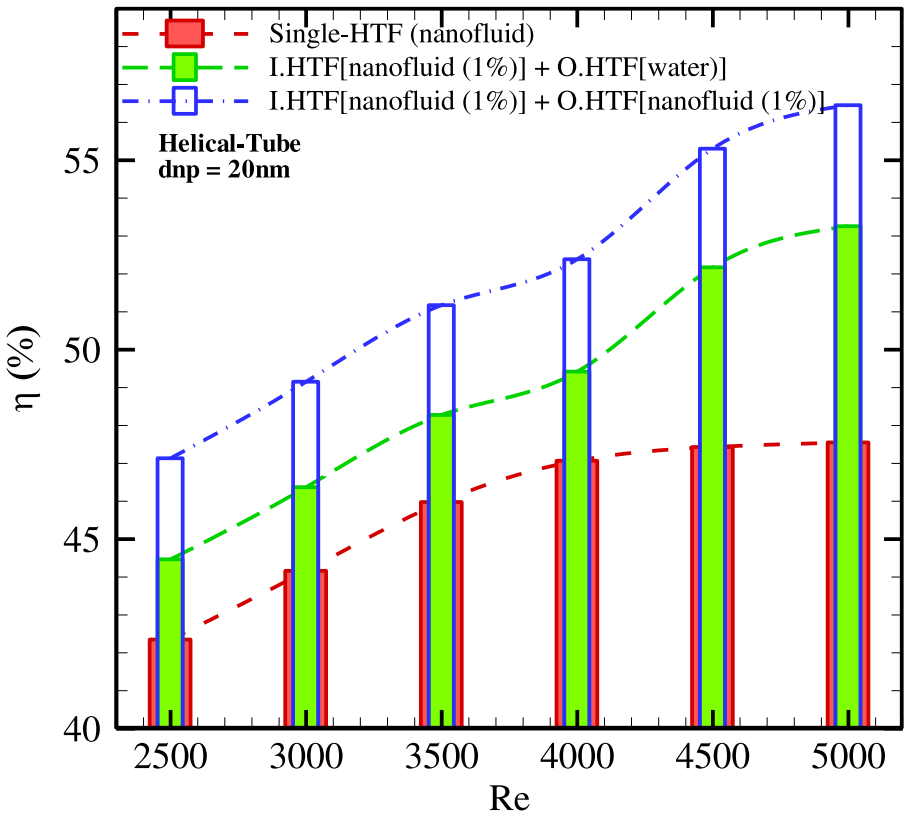


Fig. 9. Variations of energy efficiency values versus different *Re* for single- and two-fluid PTSCs equipped with helical absorber tube system filled with nanofluid.

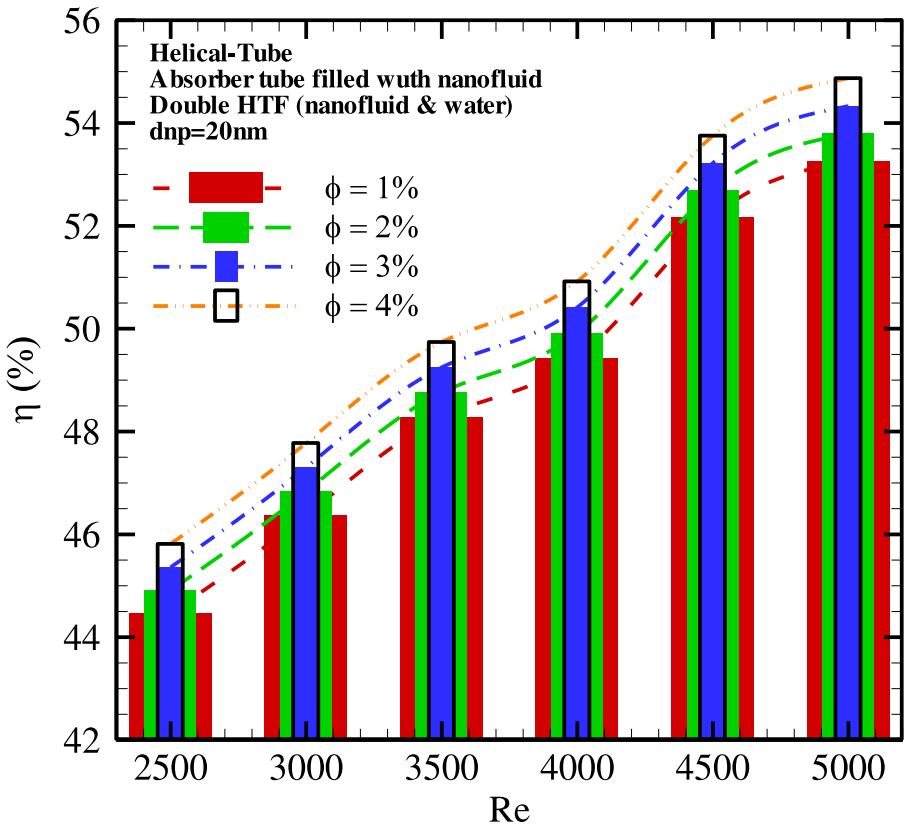


Fig. 10. Variations of the energy efficiency of a double-fluid PTSC versus different *Re* prepared with helical tube system occupied by nanofluid and glass cover annulus occupied with base fluid for various ϕ at constant diameter.

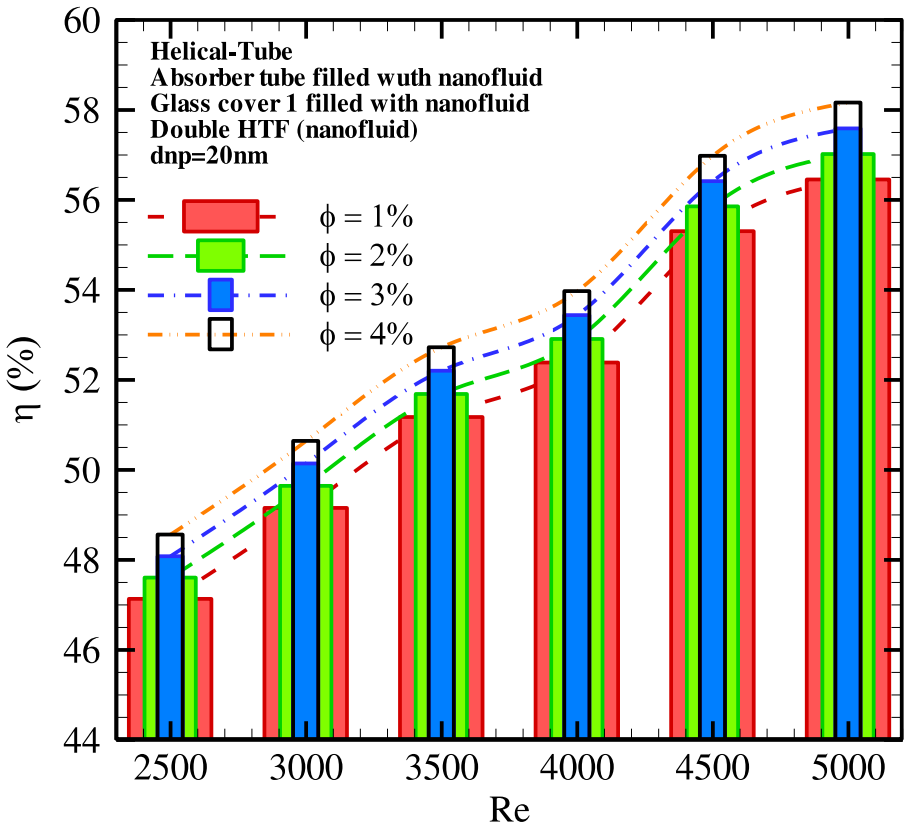


Fig. 11. Variations of first law efficiencies of a double-flow PTSC versus different flow velocities equipped with the helical absorber tube scheme filled with nanofluid at various ϕ and constant diameter (for internal fluid).

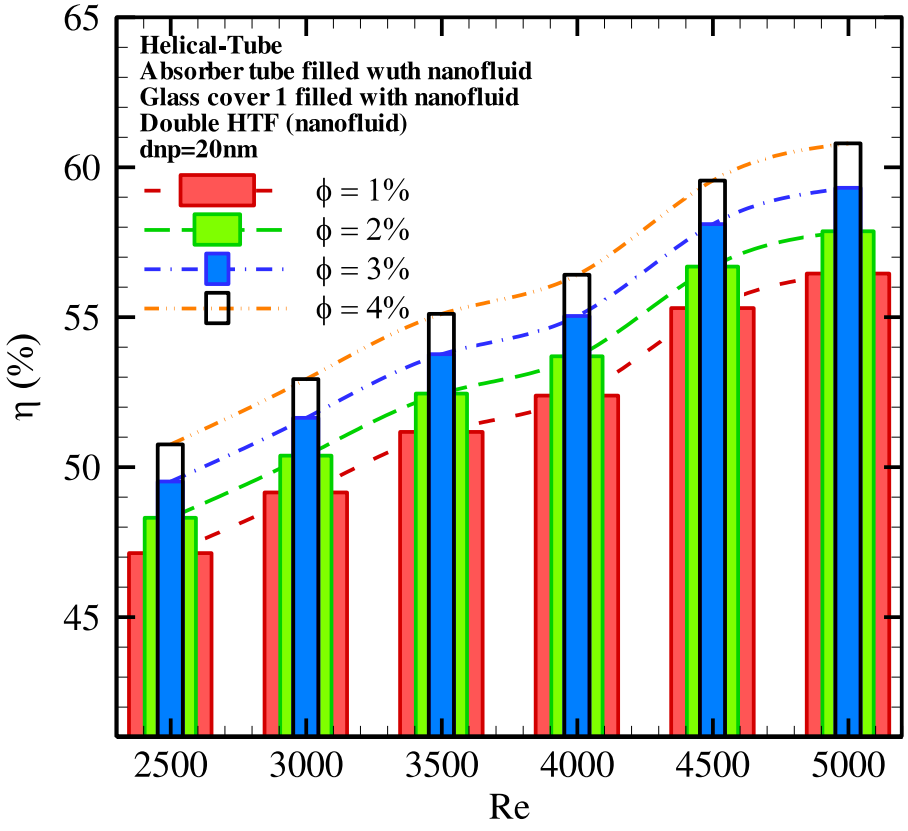


Fig. 12. Variations of energy efficiency of a double-fluid PTSC Vs. different *Re* arranged with helical absorber tube arrangement occupied by nanofluid at various nanoparticles ϕ with constant diameter (for external fluid).

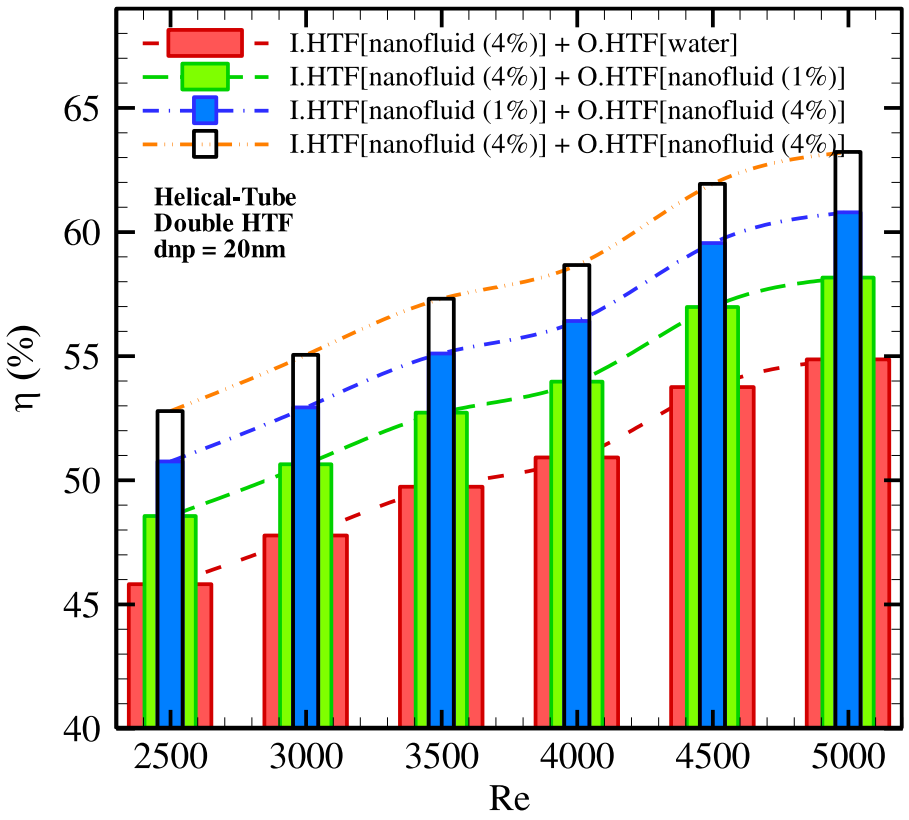


Fig. 13. Variations of first law efficiency of a double-flow PTSC versus different flow velocities equipped with helical tube system filled with nanofluid at various nanoparticles volume concentrations for optimum cases.

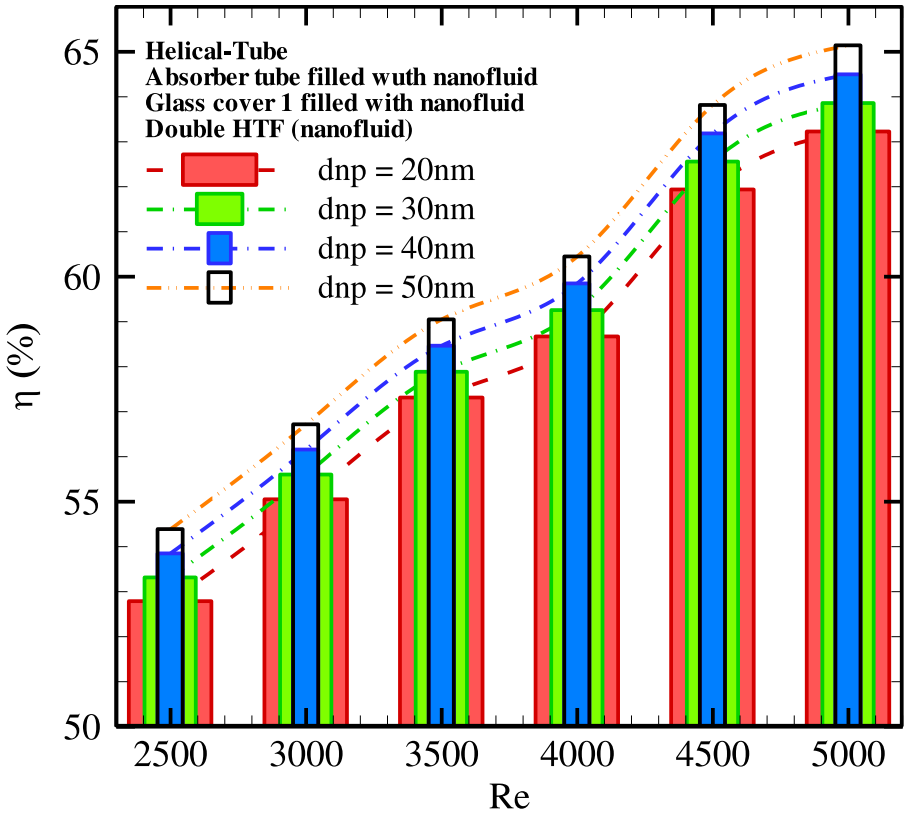


Fig. 14. Variations of energy efficiency of a double-fluid PTSC versus different Re arranged with helical absorber tube system occupied with nanofluid at various nanoparticles diameters.

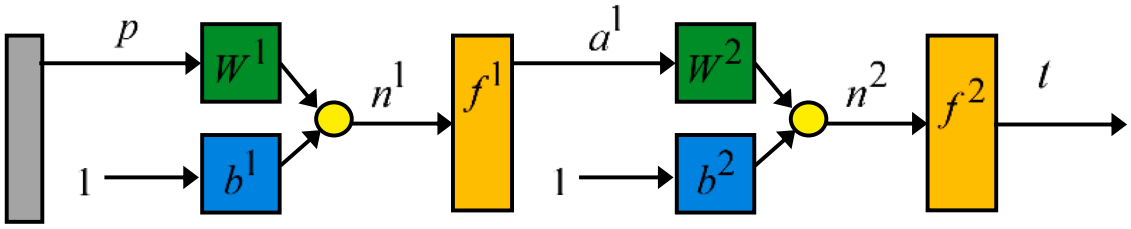


Fig. 15. A schematic view from the employed architecture.

external fluid). It is clearly understood from Fig. 12 that occupied nanofluid with different nanoparticles volume fractions has a significant impact on the energy efficiency enhancement of PTSC. It can be realized that the energy efficiency increases with the increasing nanoparticles volume fractions. The PTSC occupied with nanofluid at $\phi = 4\%$ has the maximum energy efficiency values among all considered cases and is followed by models filled with nanofluid at $\phi = 3$, 2 and 1%, respectively. Increasing volume fractions of nanofluids increase thermal conductivity and dynamic viscosity of nanofluids into HTF. Therefore $h_{convection}$ between HTF and tube enhanced by the increasing thermal conductivity of nanofluid. Besides, more dynamic viscosity in HTF increases the pressure drop between inlet and outlet sections. Also, it is seen that for all considered models in this figure, the energy efficiency values are enhanced by an increase of *Re*. Therefore, it can be assumed that there will be η_{max} on $V_{optimum}$. The value of η_{max} at 1 and 4 vol.% is 55.6% and 60.4%, respectively, at the flow velocity of Re = 5000.

Fig. 13 demonstrates variations of first law efficiencies of a double-flow PTSC versus various *Re* arranged with helical tube system occupied with nanofluid at various volume concentrations for the optimum cases. Fig. 13 illustrates the first law efficiency changes during an increase of studied flow velocities for the double-flow PTSC with a helical tube filled with 4% nanofluid in the absorber tube and base fluid in the internal annulus, 4% nanofluid in the absorbent channel and 1vol% nanofluid in the annulus, $\phi = 1$vol% in the absorbent channel and $\phi = 4\%$ in glass annulus and $\phi = 4$vol% in the absorber tube and $\phi = 4$vol% in the glass annulus. The nanoparticles diameter is 20nm. As it is

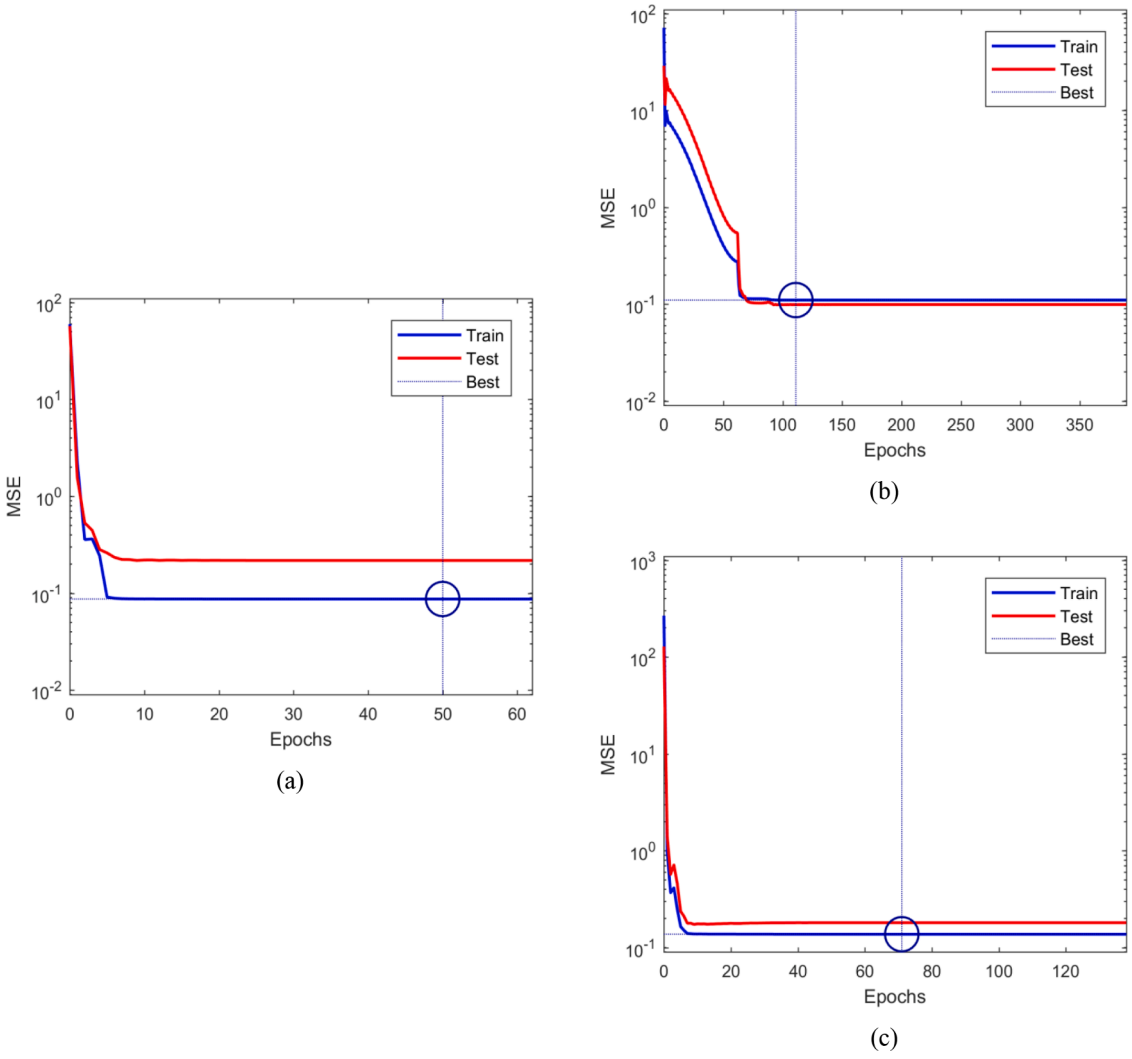
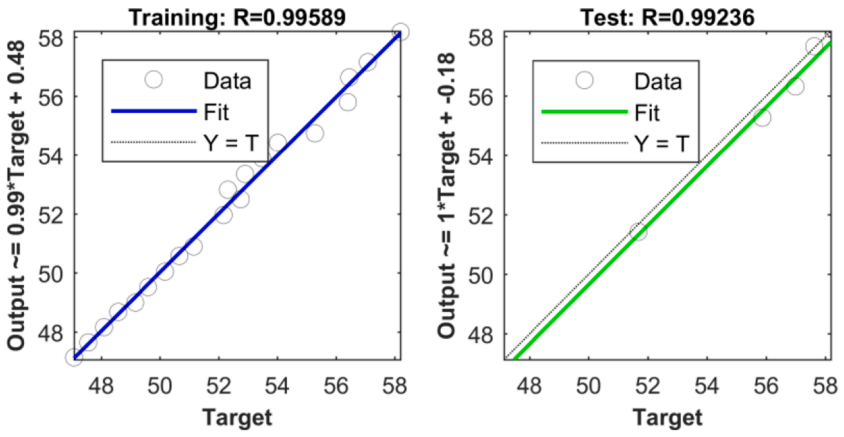
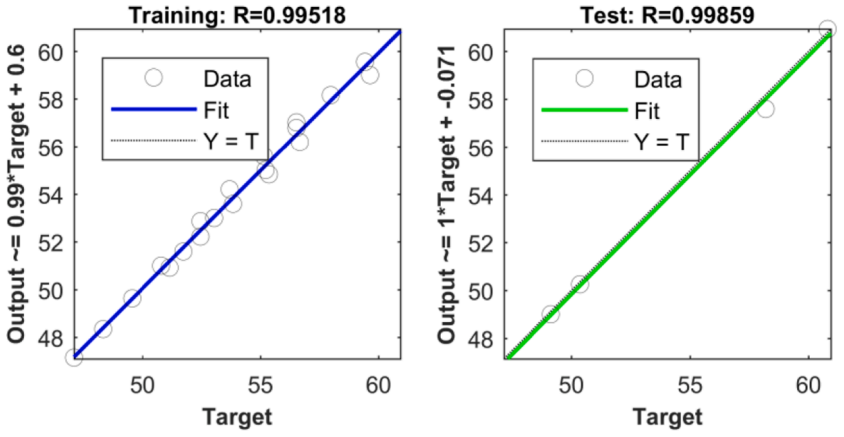


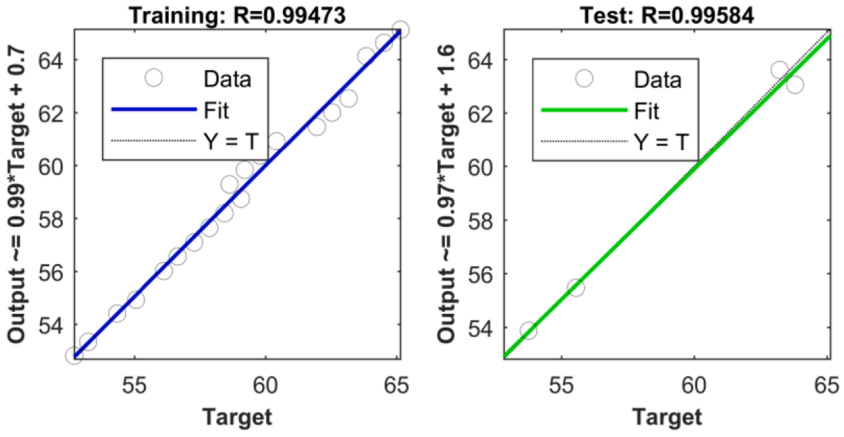
Fig. 16. The training performance of the ANN for: (a) case 1, (b) case 2, and (c) case 3.



(a)



(b)



(c)

Fig. 17. The regression plots of the ANN models for: (a) case 1, (b) case 2, and (c) case 3.

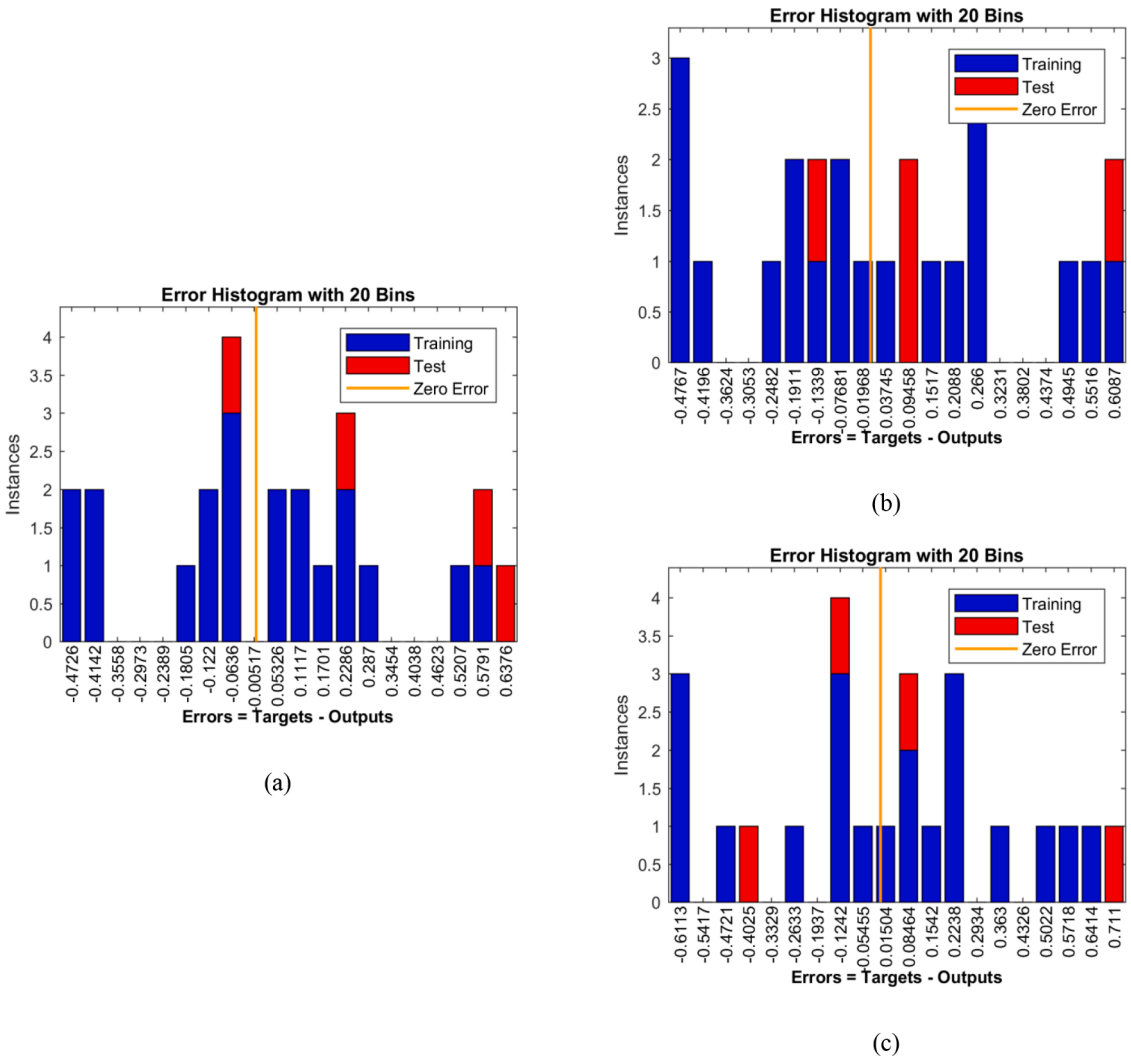


Fig. 18. The error histograms of the ANN models for: (a) case 1, (b) case 2, and (c) case 3.

realized in this figure, for all four cases, the energy efficiency was augmented by increasing*Re*. Also, it is found that during all studied flow velocities, the first law efficiency from high to low resembles to 4% nanofluid in the absorber tube and 4% nanofluid in the glass cover, 1% nanofluid in absorber tube and 4% nanofluid in the glass cover, $\phi = 4$vol % in absorber tube and 1% nanofluids in the glass cover and 4% nanofluid in absorber tube and water in the glass cover. The maximum energy efficiency of 63.2% is obtained for ϕ 4vol% in the absorber tube, and 4% nanofluid in the glass cover and the one for $\phi = 4$ vol% in the absorber tube and water outside is 54.9% at Re = 5000.

Fig. 14 shows variations of the first law efficiency of a double-flow PTSC versus different flow velocities equipped with a helical absorber tube system filled with nanofluid at various nanoparticles diameters. Fig. 14 illustrates the changes in energy efficiency terms versus different turbulent Reynolds numbers for nanoparticles at various sizes of 50nm, 40nm, 30nm and 20nm for a double-fluid PTSC with a helical tube filled with suspension with constant volume concentrations of 4% in the absorbent channel and annulus. The purpose is to study the effects of nanoparticles diameters on the energy efficiency of PTSC. For all four different diameters and the considered Re, the energy efficiency always upsurge. Similarly, it is understood that during all studied flow velocities, the energy efficiency from high to low matches to sizes of 50nm, 40nm, 30nm and 20nm. The maximum energy efficiency is 65.1% and 63.1% for the diameters of 50nm and 20nm, respectively, at the flow velocity of Re = 5000.

Finally, it has been understood that the double-flow PTSC equipped with helical absorber tube occupied with $\phi = 4\%$ nanofluid and nanoparticles diameter of 50nm flowing in the absorbent channel and the glass annulus is presented as the best model in this paper.

4. Deep Learning Model

As mentioned in the introduction, artificial intelligence techniques, especially machine learning, can help reduce costs and time to solve problems. Many researchers have researched this area and reported satisfactory results [93–95]. In this section, a deep learning method is employed to examine the parameters affecting the efficiency of the double-fluid parabolic trough solar collector occupied by non-Newtonian nanofluid. These parameters include the concentration and diameter of nanoparticles as well as Re number. To this end, an Artificial Neural Network (ANN) model is utilized. The ANN model has a two-layer feed-forward architecture: one hidden layer with a “tansig” transfer function and one output layer with “purelin” transfer function. The number of hidden neurons is selected by trial and error. In this method, all possible numbers of hidden neurons are examined, and the Mean Squared Error (MSE) between the ANN outputs and targets are found. Since the weight and biases of the ANN model are randomly selected, for each number of hidden neurons, the ANN is trained 10 times. The average MSE should be considered. The number of hidden neurons corresponding to the minimized average MSE should be

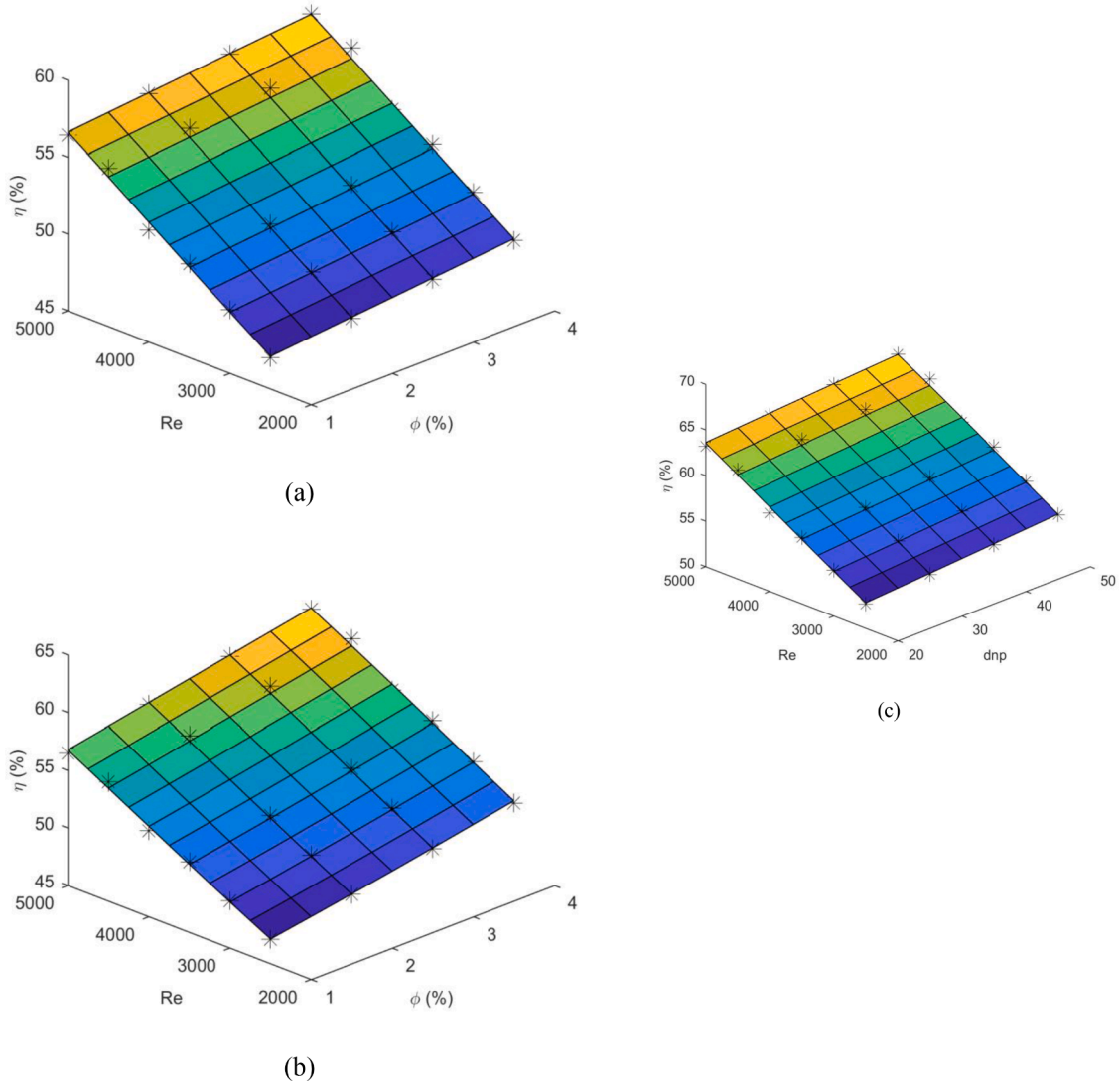


Fig. 19. The ANN results for non-trained input pairs in: (a) case 1, (b) case 2, and (c) case 3.

selected. In this paper, the number of hidden neurons is 20. There are 2 inputs and 1 output for the model. A schematic view of the employed architecture is provided in Fig. 15.

This structure is represented as follows:

$$\begin{aligned} t &= f^2(n^2) \\ n^2 &= W^2 a^1 + b^2 \\ a^1 &= f^1(n^1) \\ n^1 &= W^1 p + b^1 \end{aligned} \tag{22}$$

in which p and t are the input and target vectors, respectively. Also, W and b are the weight and bias matrices, and the superscripts show the layer number.

There are 3 sets of input-target pairs with 24 samples to investigate:

- Case 1: Variations of efficiencies due to different nanoparticle concentrations and Re numbers for internal fluid
- Case 2: Variations of efficiencies due to different nanoparticle concentrations and Re numbers for external fluid
- Case 3: Variations of efficiencies due to different nanoparticle diameters and Re numbers

The Bayesian regularization backpropagation learning algorithm is used for tuning the weight and bias values so that the MS) between the outputs (i.e., the efficiency values predicted by the ANN model) and the targets (i.e., the efficiency values obtained by the numerical simulations) is minimized. Data sets are divided randomly into "training" and "test" sets to construct and evaluate the model. While 83% of input-target samples are used for training, the remainder is utilized for test.

The training performance of the ANN is illustrated in Fig. 16. As can be seen, the MSE is converged to a minimum for both the training and test datasets during the training iterations (i.e., Epochs). Therefore, one can conclude that the ANN performance is acceptable from the training point of view.

Also, the regression plots of the ANN models are shown in Fig. 17 in which the targets (i.e., the efficiency values obtained by the numerical simulations) are plotted against the outputs (i.e., the efficiency values predicted by the ANN model) for both the training and test datasets. It can be seen that the ANN models predict outputs that are very similar to the targets. The small regression values and the similarity of the fitted lines with the target=output lines confirm that the ANN model can perfectly predict the efficiency values.

To investigate the errors, one may need to error histograms. The error histograms of the ANN models are shown in Fig. 18 in which the instances of errors are plotted against the error values within 20 bins for the training and test cases. It can be observed that the errors are

distributed near the zero-error lines. Therefore, the ANN models have acceptable error distributions. For example, the maximum efficiency error is 0.711 % throughout the ANN results, while the error values are usually smaller.

Finally, the ANN results for non-trained input pairs are illustrated in Fig. 19. The efficiency values are plotted for different input pairs in these contours, while asterisks show the targets. The results indicate the ANN has acceptable performance in predicting the efficiency of the double-fluid parabolic trough solar collector occupied by non-Newtonian nanofluid.

5. Conclusions

The present study considered the influence of using a helical absorber tube on the performances of double-fluid PTSC filled with hybrid non-Newtonian nanofluids. The obtained data in the current study can be listed as follow:

- For both conventional and helical tubes, the thermal efficiency has always been increased by increasing the flow velocities.
- The optimum flow velocity with η_{max} is related to Reynolds number of 5000.
- The thermal efficiency of helical absorber tube is during all studied flow velocities greater than the thermal efficiency of the basic absorber tube.
- For the case of helical system, the tube diameter decreases assuming the same hydraulic diameters, which increases the thermal diffusion within the fluid flow. It is the cause of first law efficiency increase in PTSC.
- The flow rate of the helical system is higher than the conventional one (at the same flow velocity). Therefore, the heat transfer rate from channel walls to HTS upsurges.
- The depth of thermal diffusion inside the HTF flow reduces for larger diameters of the absorber tube. Also, the presence of helical-shaped sections in the flow path causes local turbulence inside the HTF flow, causing in laminar sub-layer breaking and increasing the heat transfer rate inside the helical tube.
- The radiation received is concentrated only on the lower part of the conventional tube, while it is concentrated on the lower and side parts of the absorbent channel in the helical case because of the absorber tube geometry.
- The double-fluid PTSC arranged with a helical absorber tube occupied with $\phi = 4\%$ and nanoparticles' diameter of 50 nm nanofluid flowing in the absorber tube and a glass cover has been presented as the optimum model.
- For single-flow PTSC, the value of η_{max} for nanofluid- and water-based PTSCs are 47.5% and 43.1%, respectively, at flow velocity of $Re = 5000$.
- In double-flow PTSC, the value of η_{max} at 1 and 4 vol.% is 56.5% and 58.2%, respectively, at the flow velocity of $Re = 5000$.
- The ANN has acceptable performance in prediction of the efficiency of the double-fluid parabolic trough solar collector occupied by non-Newtonian nanofluid.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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